Class Meeting Handout #12

Barry Linkletter

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*, Diagrams were edited in *Affinity Publisher*

Some ideas and problems to help guide our discussion as we introduce the concept of linear free relationship.

Linear Free Energy Relationships. Part 1.

It follows that changing the electron density near a charged group would change the energy of the system.[1] One example is the effect of substituents on acid equilibrium and reaction rates. See the pK_a values of substituted benzoic acids and the rate constants for base hydrolysis of the corresponding esters as outlined in tables 1 and 2.

[1] For a discussion of substituent effects see ch. 8.2 and 8.3

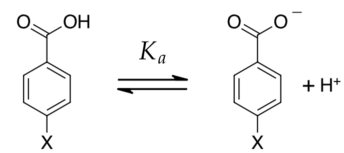

Figure 1: Acid equilibrium for benzoic acid

Exercise: The Hammett Plot – 20 minutes

Can we use one reaction to measure another? Let us see.

1. Examine table 1, which lists rate data for base-catalyzed hydrolysis of substituted benzoate esters, and use it to establish which substituents are electron-donating and withdrawing. Are there any surprises?

2. We need a numeric score for the electron-withdrawing/donating ability of a substituent. How did Hammett approach this problem?

3. Examine table 2, which lists rate data for base-catalyzed hydrolysis of substituted benzoate esters. Plot the rate of base-catalyzed hydrolysis of a series of substituted esters of benzoic acid vs. the pK_a value for the corresponding benzoic acid (table 1) and the Hammett σ value (table 3). Use a log-log plot to preserve linear relationships. Be sure to justify how the values you are using are all log values.[2]

4. Examine the plot. Is there a correlation? Do you think that the rate of hydrolysis is more or less sensitive to electronic effects compared to the acid equilibria?

5. Draw out the stepwise mechanism for base-catalyzed hydrolysis of ethyl benzoate and use this to explain the observation above.

6. Relate the line equation for the plot to the parameters of the Hammett equation.

7. Show that the Hammett equation expresses that two systems are directly correlated in the changes in free energy caused by the substituent effects.[3]

Table 1: K_a values for substituted benzoic acids.

Substituent	K_a /10^{-5}	pK_a
p-NO_2	35.5	3.45
p-CN	27.5	3.56
p-Cl	10	4.0
p-H	6.3	4.20
p-CH_3	4.3	4.36
p-OCH_3	3.4	4.47

[2] Is the Hammett σ series a logarithmic scale?

Table 2: Rate constants for hydrolysis of ethyl benzoates.

Substituent	k /mol L^{-1} s^{-1}	$\log k$
p-NO_2	32.9	1.52
p-CN	15.7	1.20
p-Cl	2.10	0.322
H	0.289	−0.529
p-CH_3	0.172	−0.764
p-OCH_3	0.143	−0.845

[3] This is the L in LFER. Ch. 8.6 will provide some help for this question.

Louis Hammett used the equilibrium of benzoic acids to create the Hammett substituent constants, the σ values.[4] Over the years many other contributors studying other systems and substituents added and tweaked values until a list was generally agreed upon.[5]

[4] "The Effect of Structure upon the Reactions of Organic Compounds. Benzene Derivatives." L.P. Hammett, *J. Am. Chem. Soc.*, **1937**, *59*, 96–103. https://doi.org/10.1021/ja01280a022

[5] See table 8.2 of the textbook. For a comprehensive list see "A survey of Hammett substituent constants and resonance and field parameters." C. Hansch, A. Leo, R.W. Taft, *Chemical Reviews*, **1991**, *91*, 165–195. https://doi.org/10.1021/cr00002a004

Exercise: Using the Hammett Equation – 20 minutes

Comparing one reaction with another can work as long as we always use the same reaction as the measuring stick. Comparing effect of substituents on benzoic acid equilibrium and in reactions involving a similar substituted benzene group can reveal the natute of the transition state.

$$\log \frac{k_X}{k_H} = \rho\sigma$$

Figure 2: The Hammett equation

8. Consider the S_N2 hydrolysis of benzyl chloride in basic and acidic conditions.
   (a) Draw the transition state for the r.d.s. of this reaction. Draw the More O'Ferrall-Jencks plot that describes the reaction coordinate for the r.d.s. Locate the t.s. on the More O'Ferrall-Jencks plot.
   (b) Using the rate data for hydrolysis presented in table 4, create a Hammett plot for the hydrolysis in hydroxide solution. Does this agree with your hypothesis for your model for the t.s.?[6]
   (c) Using the rate data for hydrolysis presented in table 4 or table 5, create a Hammett plot for the hydrolysis in acidic solution.[7] Does this agree with your model for the transition state?

[6] What does the sign and the magnitude of the Hammett reaction constant, ρ, tell you about the nature of the transition state?

[7] The authors observed that the rate of hydrolysis was constant in various concentrations of acid.

9. Consider the S_N1 solvolysis of substituted diphenylmethylenechlorides and triphenylmethyl acetates.
   (a) Draw the transition state for the r.d.s. of this reaction. Draw the More O'Ferrall-Jencks plot that describes the reaction coordinate for the reaction. Locate the t.s. on the More O'Ferrall-Jencks plot.
   (b) Using the rate data for hydrolysis presented in tables 6 and 7, create Hammett plots for the solvolysis reactions. Use these results to confirm or refute the location of the t.s. in your More O'Ferrall-Jencks plot.
   (c) Do these plots seem to have the quality of the line-fit compared to the cases of the S_N2 reactions above? Can you propose why there is more scatter here.[8]

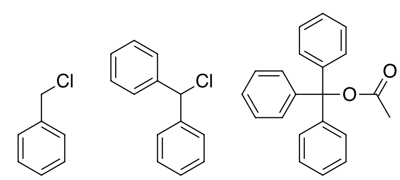

Figure 3: Benzyl chloride, diphenylmethylenechloride and triphenylmethyl acetate

[8] The work was from the 1930's, yet our ancestors dressed better than we do today and performed their science very carefully. So don't blame the workmanship; be sure to blame the molecules.

10. In the class meeting you may have observed that I ignored the ortho-substituted molecules in the data sets. Was I being too picky or was I justified in doing so?

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- have reviewed the concepts of reaction coordinates, transition states and More O'Ferrall-Jencks plots.
- have reviewed how substituents affect the energy of high-energy intermediates such as carbocations and anions.
- have considered how substituents affect the energy of ground start molecules such as carboxylate or ammonium groups.
- understand how Hammett used the benzoic acid equilibrium as the standard to create a scoring system to determine substituent constants that are directly correlated to free energy differences.
- be able to apply the Hammett equation to determine the reaction constant and be able to interpret the nature of the intermediate or transition state in a reaction using this value.
- **Computer Skill**: be able to use an interactive *Python* notebook to import, clean and manipulate date and then perform and document curve fits of the data to mathematical models.

Table 3: Hammett σ values for selected substituents.

Substituent	σ
m-Br	0.39
p-Br	0.23
m-CH_3	−0.06
p-CH_3	−0.14
m-C_6H_5	0.06
p-C_6H_5	−0.01
m-Cl	0.37
p-Cl	0.23
m-CN	0.56
p-CN	0.66
m-I	0.35
p-I	0.18
m-$N(CH_3)_2$	−0.15
p-$N(CH_3)_2$	−0.83
m-NO_2	0.72
p-NO_2	0.78
m-OCH_3	0.11
p-OCH_3	−0.27
m-OC_6H_5	0.25
p-OC_6H_5	−0.32
m-OEt	0.1
p-OEt	−0.24

Values in table 3 were obtained from "Free Energy Relationships in Organic and Bioorganic Chemistry" by Andrew Williams, *The Royal Society of Chemistry, Cambridge*, **2003**, pp 258-277. https://doi.org/10.1039/9781847550927. We have access to the E-textbook version via the UPEI library. Other sources of σ values are described in reference 5

A Look Ahead

Some of the plots seemed lower in quality than others. Is there a pattern? Were there particular substituents that were more likely to fall off the line? I trust the researchers' measurements. Is it the Hammett parameters that are wrong? I also have faith in Louis Hammett.

We shall see that Hammett's σ parameters don't capture all of the effect that some substituents can provide. Hammett's reference system of benzoic acid dissociation was not sensitive to resonance effects and some of the reactions discussed today are very sensitive to resonance effects from some of the substituents. We will need a new set of parameters.[9] We will explore this idea in the next class meeting.

[9] See ch. 8.3.5 of the textbook

Comments and Analysis

Many plots will be presented today using *Python* notebooks. It will happen fast and so I have prepared a document containing the plots and commentary that will be available on the course moodle page. Look for it there.

Data Tables

	$[H_2SO_4]$	[NaOH] / mol L^{-1}		
	0.27	0.051	0.104	0.259
Substituent	k_{obs} / 10^{-5} min^{-1}			
H	2.2	17	31	61
p-CH_3	19	38	58	94
o-CH_3	9.4	42	68	127
m-CH_3	2.6	33.2	28	53
p-Br	1.09	17	34	64
o-Br	0.58	11	22	47
m-Br	0.41	9.0	17	37
m-CN	0.27	17	33	68
p-CN	0.25	22	43	92
o-CN	0.21	24	46	102

Table 4: Hydrolysis of benzylchloride in 50% acetone/water in the presence of acid and base. Data from "Relations Entre la Constitution du Substrat et sa Sensibilité à L'ion oh dans L'hydrolyse." S.C.J. Olivier, A.P. Weber, *Recl. Trav. Chim. Pays-Bas*, **1934**, *53*, 869–890. https://doi.org/10.1002/recl.19340531002 All experiments were at 30 °C

	48 % EtOH		50 % Acetone	
	30 °C	83 °C	30 °C	60 °C
Substituent	k_{obs} / 10^{-3} min^{-1}			
H	0.111	15.5	0.0225	0.46
p-CH_3	1.04	164	0.173	4.0
o-CH_3	0.55	75	0.098	1.89
m-CH_3	0.144	21.6	0.0278	0.55
p-Cl	0.052	9.6	0.0126	0.268
o-Cl		5.5	0.0061	0.137
m-Cl	0.0152	3.68	0.003 99	0.093
p-Br	0.0457	7.8	0.0110	0.234
o-Br	0.0236	4.44	0.0055	0.129
m-Br	0.0147	3.34	0.003 90	0.093
p-I	0.0414	7.4	0.0100	0.215
o-I	0.0248	4.45	0.0058	0.123
m-I	0.0152	3.11	0.003 96	0.089
m-COOH	0.0189	3.79	0.0055	0.132
p-COOH	0.0121	2.56	0.003 91	0.086
m-NO_2	0.0063	1.40	0.002 13	0.051
o-NO_2	0.0052	1.30	0.001 62	0.045
p-NO_2	0.004 91	1.15	0.001 97	0.046

Table 5: Solvolysis of benzylchloride in solvent/water. Data from "Sur un Parallélisme Entre la Mobilité de L'hydrogène du Noyau Benzénique et Celle du Chlore de la Chaîne Latérale." S.C.J. Olivier, *Recl. Trav. Chim. Pays-Bas*, **1930**, *49*, 697–704. https://doi.org/10.1002/recl.19300490803

I did not find this data myself. I used references in the comprehensive review by Jaffé: "A Reexamination of the Hammett Equation", H.H. Jaffe, *Chem. Rev.*, **1953**, *53*, 191–261. https://doi.org/10.1021/cr60165a003. Often, following a chain of references in the literature is more useful than taking the first thing that Google coughs up or what an LLM system imagines.

Ring A		Ring B	Methanol	Isopropanol	Ethanol	
Substituents			k_{obs} / 10^{-3}min^{-1}			
H	H	H	48.9	0.343	2.66	
o-Cl	H	H	0.615		0.0254	
m-Cl	H	H	1.27			
p-Cl	H	H	22.8	0.175	1.07	
p-CH_3	H	H			43.3	
o-OCH_3	H	H				
m-OEt	H	H		0.396		
p-OCH_3	H	H		1700		
p-NO_2	H	H	0.0330			
m-Cl	p-Cl	H	0.902			
p-Cl	H	p-Cl	6.27	0.057	0.40	
o-Cl	p-Cl	p-Cl	0.113			
p-CH_3	H	p-Cl		2.74		
p-OCH_3	m-Cl	H		48.4		
p-OCH_3	m-Cl	p-Cl		24.1		
p-C_6H_5	H	H			3.62	
m-Cl	H	H			0.059	*
p-Br	H	H			0.883	*
m-CH_3	H	H			5.54	*
o-CH_3	H	H			7.64	*
p-Et	H	H			5.58	*
o-OCH_3	H	H			247	*
p-OC_6H_5	H	H			843	*
p-CH_3	H.	p-CH_3			1100	*

Table 6: Solvolysis of diphenylmethylenechlorides. Data for methanol and isopropanol from S. Altscher, R. Baltzly, S.W. Blackman, *J. Am. Chem. Soc.*, **1952**, *74*, 3649–3652. https://doi.org/10.1021/ja01134a054. Data for ethanol from EtOH results from J.F. Norris, C. Banta *J. Am. Chem. Soc.*, **1928**, *50*, 1804–1808. https://doi.org/10.1021/ja01393a049. Additional data (*) from J.F. Norris, J.T. Blake, *J. Am. Chem. Soc.*, **1928**, *50*, 1808–1812. https://doi.org/10.1021/ja01393a050. All experiments were at 25 °C

Substituents			pK_r	90 % CH_3CN	80 %	60 %	50 %
				k_{obs} / 10^{-3}s^{-1}			
H	H	H	−6.63	0.0147	0.0588	0.270	0.557
p-Me	H	H	−5.41	0.103	0.359	1.46	3.01
p-Me	p-Me	H	−4.71	0.323	1.21	5.62	9.59
p-Me	p-Me	p-Me	−3.56	1.30	4.98	17.7	33.3
p-OMe	H	H	−3.40	1.20	4.53	15.0	24.0
p-OMe	p-OMe	H	−1.24	40.4	115	306	441
p-OMe	p-OMe	p-OMe	0.82	680	1580	3860	5560
p-$N(CH_3)_2$	H	H	3.88	1080	2000	4510	7400
p-$N(CH_3)_2$	p-OMe	H	4.86	6230	12 200	24 900	39 300
p-$N(CH_3)_2$	p-$N(CH_3)_2$	H	6.94	128 000	215 000		

Table 7: Solvolysis of triphenylmethylacetates in water/acetonitrile. Data from "Stabilities of Trityl-Protected Substrates: The Wide Mechanistic Spectrum of Trityl Ester Hydrolyses." M. Horn, H. Mayr, *Chem. Eur. J.*, **2010**, *16*, 7469-7477. https://doi.org/10.1002/chem.200902669. All experiments were at 25 °C